Mathematics in the Modern World

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The Symbolic Language of Mathematics

- Mathematics is written in a **symbolic language** that is designed to express mathematical thoughts.
- English language is a source of knowledge, but it is not designed for doing mathematics.

The Symbolic Language of Mathematics

- This chapter describes how mathematical thoughts, methods, and facts are expressed in symbolism.
- That is, it emphasizes how to read and write mathematics.

Mathematical English:

- Mathematical English is a part of the English language used for making formal mathematical statements, specifically to communicate definitions, theorems, proofs and examples.

- Many ordinary English words are used in Mathematical English with different meanings. Other languages also have special mathematical forms.

Symbolic Language:

- The symbolic language of mathematics is a special-purpose language.
- It has its own symbols and rules of grammar that are quite different from those of English.
- We can usually read expressions in the symbolic language in any Mathematical article written in any language.

Symbolic Language:

- Both the languages, Mathematical English and the Symbolic language are used in mathematics writing and mathematical lectures.

Translating Verbal Phrases

Objective:

To recognize symbols, variables, and types of sentences used in algebra.

Introduction

- The ability to translate verbal phrases has real-life application…

- Suppose you wanted to throw a party and you only have P15,000 to work with…you contact several catering companies and they give you prices ranging from P375.00 to P500.00 per person…you also have to buy decorations with the P15,000.00.

Introduction

- In real life you will need to be able to calculate the total cost to know that you have enough money to pay for the party…Which will tell you how many friends you can invite…

- In this instance you can create an algebraic expression to know the number of friends you can invite and to make sure that you stay within your budget.

Translating Verbal Phrases

- The key to translating verbal phrases it to know what the English words mean mathematically…
- It's expected that you know the words that mean add, subtract, multiply and divide
- Let's do a quick review to refresh your memory…

Math Symbols

*	Asterisk	÷	Division
[ ]	Brackets	√a	Square root
a^b	Caret	3√	Cube root
=	Equality	π	Pi
≠	Not equality	!	Exclamation mark
≈	Approximately equal	/	Division slash
<	Less than	;	Semicolon
>	Greater than	%	Percentage
( )	Parentheses	&	Ampersand
+	Addition	∞	Infinity
—	Subtraction	$	Dollar sign

Equalities

=	Equals- is the same as
≅	Congruent- same size and shape
~	Similar- same shape

Inequalities

<	Is less than
>	Is greater than
≤	Is less than or equal to
≈	Approx. equal to
≠	Not equal to

Expressions vs. Equations

Sentences

	Expressions	Equations	Inequalities
Numerical	2 + 3 5(8) - 4	2 + 3 = 5 4 + 2(3) = 10	9 - 5 > 3
Variable	x + 7 8 - 3y	x - 4 = 13 11 = 3 + 2m	6y - 4 < 8

Open sentences

Open sentences have solutions and can be solved.

Identify each as an expression, sentence, open sentence, equation, or inequality.

1) **3x + 5 = 11**

Sentence, open sentence, equation

2) **7 < 2(5) + 3**

Sentence, inequality

3) **5x - 2**

Expression

4) **6m + 2 > 3**

Sentence, open sentence, inequality

Open sentences have solutions and can be solved.

Variable Equations

$$4 + m = 7$$
$$-4 \qquad -4$$

$$m = 3$$

One Solution

Variable Inequalities

$$5 + y < 91$$
$$-5 \qquad -5$$

$$y < 86$$

Infinite Solutions

Give three solutions to each sentence below.

1) $x > 10$ Samples: 11, 15, 34

2) $x + 3 \leq 7$ Samples: 4, -8, 0

3) $5 - x < 0$ Samples: 6, 7, 10

4) $2x - 1 > 3$ Samples: 3, 4, 7

To translate English words, phrases, and sentences into mathematical symbols, expressions, and equations respectively.

+	−	×	÷
plus	minus	times	divide
add	subtract	multiply	quotient
increased by	decreased by	product	
sum	diminished by	…of…	
total	difference	twice (× 2)	
	less		
more than	less than subtracted from		

Key Words

Key words can help us solve mathematical word problems.



Addition

add	both	join
add to, added to	combined	more, more than
addition of	extra	plus
all together	in all	sum
altogether	increase by	together
and	increased by	total, total of



Subtraction

decrease, decreased	how much less	minus
deduct	how many more	nearer, further
difference	how much more	reduce, reduced by
difference between	how many left	remaining
dropped, fell	how much left	subtract
fewer, fewer than	less, less than	take away



Multiplication

area of	multiplied by	rows of
at	of	sets of
by	multiple	times
double, doubled	per	triple
each had	product, product of	twice
groups of	rate	



Division

divide evenly	for each	quotient
divided by	half, quarter	ratio, ratio of
equal parts	how many each	share, share of
equal pieces	out of	shared between
fraction	per	shared equally
	percent, percentage	split



Equals

answer	gives	results in, result is
are	is, is equal to	same amount
corresponds to	makes	same as
equals	produces	same value
equates to	product is	yields

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Translating Verbal Phrases

- The starting point to translate verbal phrases is to **identify the variable first**…
- Most often you will know what the variable is by the phrase *"a number"*…
- One more thing that you need to know…the Commutative Property applies to addition and multiplication…generally, ***the property states "it doesn't matter which order you add or multiply…you will get the same results"***
- However, when subtracting or dividing it **DOES** matter which order you place the numbers….

Tricky Ones!

5 plus x	5 + x
add 5 and x	5 + x
the sum of 5 and x	5 + x
5 increased by x	5 + x
5 more than x	x + 5
5 is more than x	5 > x

Tricky Ones!

5 minus x	5 - x
5 less x	5 - x
the difference of 5 and x	5 - x
5 decreased by x	5 - x
5 less than x	x - 5
5 subtracted from x	x - 5
5 is less than x	5 < x

Example # 1

Five years older than her brother

1. First identify the variable…in this case the variable is her brother's age…lets call that ***a***
2. The term "older than" means to add
3. Five years means the number 5

So the above expression can be written as:

5 + *a*

Example # 2

Six dollars an hour times the number of hours

1. Hour is the variable …let's call it h
2. Times means to multiply
3. Six dollars means the number 6

The algebraic expression is:

$$6 \cdot h$$

This can also be written as $6h$

Example # 3

Three more than the quantity five times a number

1. 5 times a number is the variable …let's call it 5n
2. More than means to add
3. Three means the number 3

The algebraic expression is:

5n + 3

Example # 4

Two less than the sum of 6 and a number *m*

1. A number ***m*** is the variable
2. The **sum** of 6 and m means to **add**
3. Two **less than** means to **subtract** 2
4. In this instance you have to add before you subtract…so the sum of 6 and m would go in parenthesis

The algebraic expression is:

(6 + *m*) – 2

Example # 5

A number *x* decreased by the sum of 10 and the square of a number y

1. A number ***x*** is the variable
2. **Decreased** means to **subtract**
3. The **sum** means to **add**
4. In this instance you have to add the sum of 10 and the square of a number y. Since you have to perform this function first before you subtract …10 and the square of y would go in parenthesis

The algebraic expression is:

$$x - (10 + y^2)$$

Example # 6

Nine less than the product of **ten and a number *d* is eleven.**

1. The variable is ***10 and a number d***, which is written as ***10d***
2. **Nine less** means to subtract 9
3. **"is"** means equal
4. The **total** is 11

The algebraic expression is:

$$10d - 9 = 11$$

- **Translate the verbal phrase into an algebraic expression. Use x for the variable in your expression.**

1. Nine more than a number
2. Three more than ½ a number
3. The quotient of a number and two tenths
4. The difference of ten and a number
5. Five squared minus a number

- **Write each sentence as an algebraic equation or inequality.**

6. Nine is greater than three times a number x.
7. Twenty-five is the quotient of a number y and 3.5.
8. Three times the quantity two less than a number x is ten.
9. The quotient of thirty-five and a number t is less than or equal to seven.
10. A number q is equal to or greater than one hundred.

Answers

1. $x + 9$
2. $\frac{1}{2} x + 3$
3. $x \div 2/10$
4. $10 - x$
5. $5^2 - x$
6. $9 > 3x$
7. $25 = y/3.5$
8. $3(x - 2) = 10$
9. $35/t \leq 7$
10. $q \geq 100$

Translate each phrase into a mathematical expression.

1) 5 subtract x

2) the product of 10 and y

3) 20 decreased by a number t

4) 6 added to the product of 11 and m

5) the quotient of y and 9

6) 8 less than 7 times k

Translate each phrase into a mathematical expression.

1) 5 subtract x **5 - x**

2) the product of 10 and y **10y**

3) 20 decreased by a number t **20 - t**

4) 6 added to the product of 11 and m **11m + 6**

5) the quotient of y and 9 $y \div 9$

6) 8 less than 7 times k **7k - 8**

Algebra Word Problem:Example 1

Josh and Jeremy bought a total of 12 apples. Josh bought 4 more apples than Jeremy. How many apples did each of them buy?

Step 1: Identify the constants and variables.

- **Constant:** total amount of apples (12).
- **Variables:** Josh bought x apples; Jeremy bought y apples.
  - (Note: x and y are commonly used placeholders for the values to be found).

Step 2: Identify the operation.

- Since the sum of the number of apples bought by each person is being considered, the operation being used is addition.

Step 4: **Obtain the solution.**

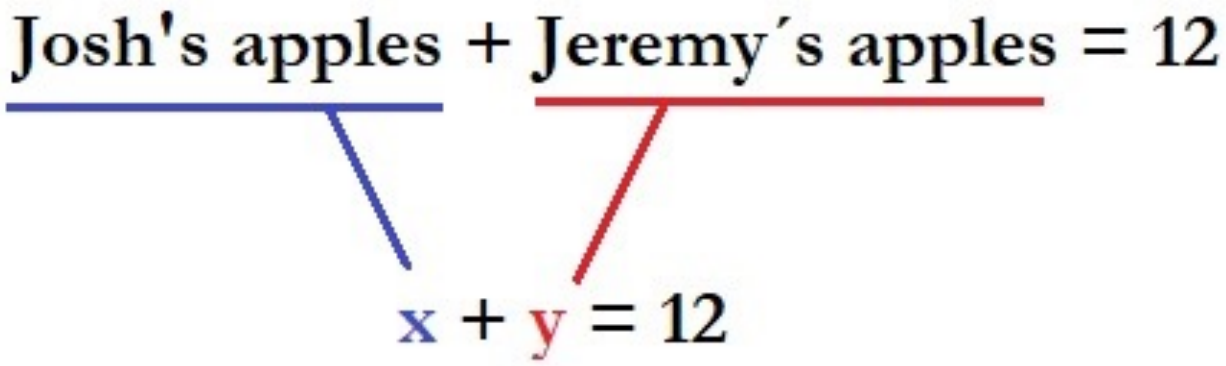
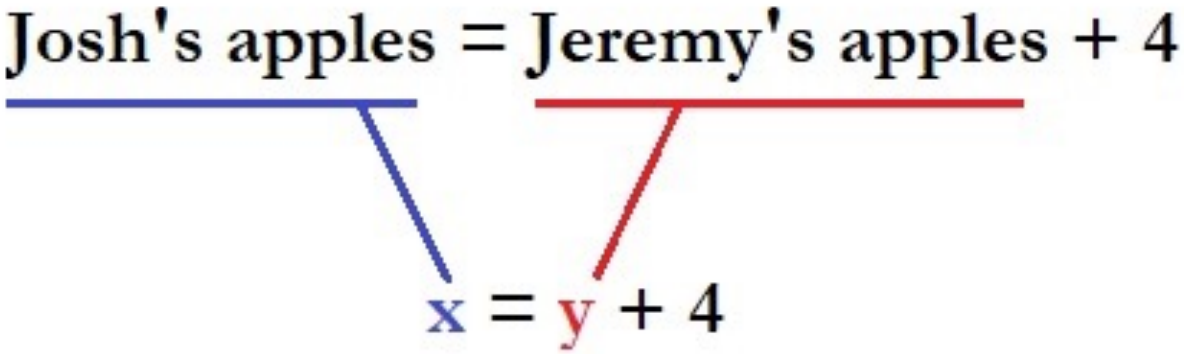


Figure 1: Josh and Jeremy bought a total of 12 apples.



Translate each sentence into a mathematical equation or inequality.

1) Eight less than twice the sum of four and y is the same as 19.

2) Six more than x is more than twice the difference of x and 7.

3) The quotient of m and 3 is six less than the product of 10 and m minus six.

4) Five less than x is less than five less x.

Translate mathematical equation or inequality into a sentence.

1) $2(4 + y) - 8 = 19$

2) $x + 6 > 2(x - 7)$

3) $m \div 3 = 10(m - 6) - 6$

4) $x - 5 < 5 - x$